Steven Zhang

(740) 278-5741 | zhang.st3ven1@gmail.com | linkedin.com/in/steven-zhang8 | GitHub | Portfolio | Columbus, OH

EDUCATION

The Ohio State University — Columbus, OH
B.S. Computer Science and Engineering, Minor in Mathematics — *April 2026*
GPA: 3.7

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), SQL, C, Ruby, MATLAB
Frameworks: Flask, FastAPI, Spring Boot 3, Spring Security, React 18, Angular 19, Pydantic, SQLAlchemy, Chart.js
Data & Infrastructure: PostgreSQL, MySQL, Redis, Supabase, Docker, Docker Compose, GitHub Actions, AWS, Linux
Tools & Testing: Git/GitHub, Bitbucket, pytest, MockMvc, Testcontainers, Flyway, Alembic, Claude Code, Codex, Wireshark, Splunk, Nmap

PROJECTS

FITT — AI-Powered Fitness & Nutrition Tracker — github.com/jmulyadi/FITT
Frontend Lead (React 18, Chart.js, FastAPI, Supabase, Groq)

- Led the frontend for a 6-person team, authoring ~3,700 lines of React/JSX across workout logging, nutrition tracking, and analytics views.
- Built a conversational AI coaching chat powered by Groq, delivering personalized training and nutrition guidance in-app.
- Defined the REST API contract with the backend team: request/response schemas and error-handling conventions.
- Contributed to the FastAPI backend, wiring ExerciseDB and Open Food Facts into routes for nutrition data.

RetireWise — Retirement Account Dashboard — github.com/BigMuscleSmoe/retirement-dashboard
Full-Stack Developer (Angular 19, Spring Boot 3, PostgreSQL, JWT, Docker) — Live Demo

- Built a Spring Boot REST API with 10 endpoints for account management, contribution tracking, balance history, and retirement projections.
- Implemented stateless JWT auth with an access/refresh token flow, BCrypt hashing, and role-based access control through the Spring Security filter chain.
- Wrote Flyway SQL migrations for schema evolution and structured JPA entities with relational mappings and Jakarta Bean Validation constraints.
- Containerized the stack with Docker Compose; set up GitHub Actions CI with build, test, and SAST scans.
- Wrote integration tests with MockMvc and Testcontainers covering endpoint behavior, auth flows, and database interactions.

URL Shortener — REST API with Analytics — github.com/BigMuscleSmoe/url-shortener
Backend Developer (Flask, PostgreSQL, Redis, Docker)

- Built a Flask REST API with three endpoints: short-link creation with URL validation, 302 redirects with async click recording, and per-link analytics.
- Implemented base62 code generation with unique-constraint collision retry, weighed against hash-based and sequential approaches.
- Applied per-IP rate limiting via Flask-Limiter backed by Redis for consistency across app instances, returning 429 on limit breach.
- Added Redis cache-aside on the redirect path with TTL expiration and negative caching for invalid codes, cutting database lookups on high-traffic links to a single cache-warming query.
- Decoupled analytics writes from the redirect path with a background thread and covered create, redirect, 404, rate-limit, and stats paths with pytest integration tests.

EXPERIENCE

The Ohio State University — Columbus, OH
Teaching Assistant / Grader — *Jan 2025 – May 2026*

- Supported 100+ students in Java and object-oriented programming through office hours and discussion boards, debugging code alongside them.
- Graded programming assignments with consistent, actionable feedback in coordination with the TA team.

Number 1 Chinese Restaurant — Steubenville, OH
Assistant Manager — *Jan 2022 – Dec 2022*

- Negotiated a zero-commission DoorDash partnership that raised monthly sales by roughly $15,000.

CERTIFICATIONS & INVOLVEMENT

- Google Cybersecurity Professional; Claude Corps; AWS Cloud Practitioner and Solutions Architect (in progress).
- Cyber Security Club, Ohio State (2024–2026) — CTFs and pentesting with Wireshark, Metasploit, Nmap.